Lecture 5

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Causal Inference via Conditional Exogeneity

Introduction

"A fancy method alone is not a credible identification strategy"

- Here we discuss how average causal effects may be identified using regression when treatment is not randomly assigned but instead depends on observed covariates.
- We discuss the regression method, where we compute the average difference between expected outcomes for treated and untreated units that are comparable (formally, identical) in terms of their characteristics X.
- If treatment is as good as randomly assigned conditional on X, then this approach recovers average causal or treatment effects.
- This key condition is commonly referred to as conditional ignorability or **conditional exogeneity**.

Example

- In a cross-country analysis, higher chocolate consumption predicts a higher number of Nobel laureates per capita.

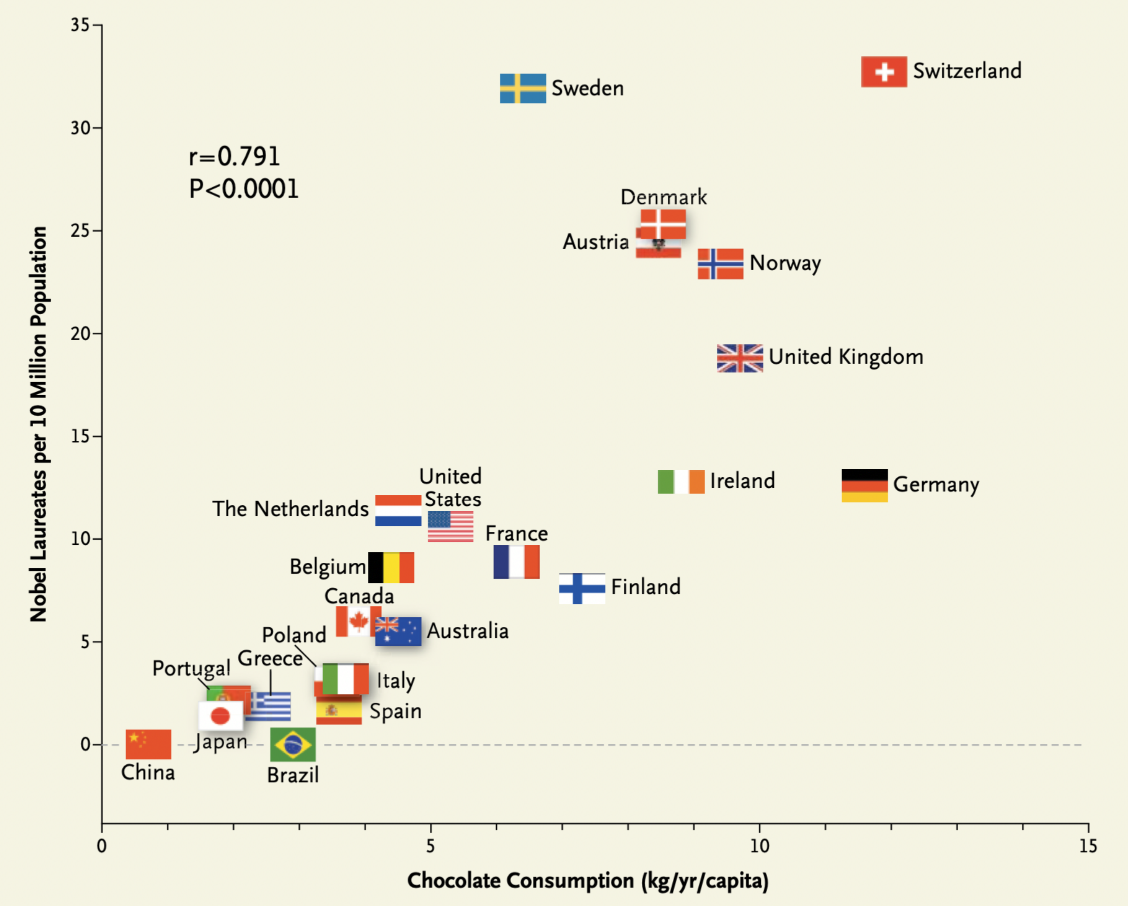
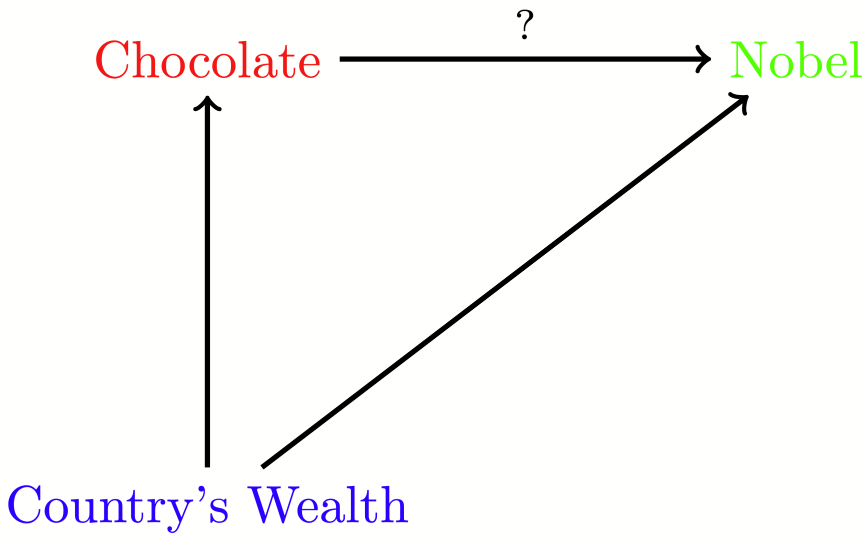


Figure 1: Source: Franz H. Messerli, "Chocolate Consumption, Cognitive Function, and Nobel Laureates", New England Journal of Medicine. 2012

- Is this a reflection of a true causal effect and therefore an actionable insight?
- If it were, countries could generate more Nobel laureates per capita by making chocolate abundant to everyone. (This wouldn't be a bad thing.)
- Is this perhaps what Switzerland did? Switzerland has the highest number of Nobel laureates per capita.

Common Causes

- Or is there a common cause that creates non-causal association – perhaps wealthy countries invest more in science and higher wealth causes people to consume luxury goods like chocolate?
- Comparative analysis, where we compare nations with identical or similar wealth, would probably reveal that the correlation is not causal.
- Probably we should be comparing Switzerland to similar countries in terms of wealth – the "apples-to-apples" comparison, so to speak.
- This type of analysis is very common in causal inference and is implemented via a set of tools introduced in this chapter.



A Contrived Causal Path Diagram for the Effect of Country's Wealth on Chocolate Consumption and Nobel Prize Production per capita.

- In what follows, we work within Rubin's potential outcomes framework(Rubin (1974)), as introduced in Lecture 2. The idea is that if we can think of observed treatment D as generated randomly – independently of potential outcomes – conditional on some pre-treatment variables X, then we can learn the average causal (treatment) effects by regression

$$\text{of } Y \text{ on } D \text{ and } X,$$

or, as is often said, by "controlling" for X.

Notation

- Recall that we denote the independence of two random variables as $\perp\!\!\!\perp$

Note that $\perp$ that is used to denote orthogonality (uncorrelatedness of a centered random variable with another)

- If U is centered, then $U \perp\!\!\!\perp V$ implies $U \perp V$, but the reverse implication is not true in general.
- Independence, conditional on a third variable X, is denoted by

$$U \perp\!\!\!\perp V \mid X.$$

Identification by Conditioning

- Recall that we use $Y(d)$ to denote potential outcome in the treatment state d.
- Suppose we want to study the impact of smoking marijuana on life longevity.
- Suppose that smoking marijuana has no causal/treatment effect on life longevity:

$$Y = Y(0) = Y(1), \text{ so that } \delta = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = 0.$$

- However, the observed smoking behavior, D, results not from an experimental study, but from observational data in which an individual's smoking behavior is associated with poor health choices X (drinking alcohol for example) which cause shorter life longevity.

- In this case, the predictive effect recovered by regression without adjusting for X does not match the average causal effect

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] < 0 = \delta,$$

because higher D predicts higher X, which predicts lower Y.

- This difference between the predictive effect and average causal effect is the result of confounding or **selection bias**.
- In this example, conditioning on X can remove the selection bias

$$\mathbb{E}[\mathbb{E}[Y \mid D = 1, X] - \mathbb{E}[Y \mid D = 0, X]] = \delta$$

provided that conditional on X variation in D is independent of the potential health outcomes.

Conditional Exogeneity

- The following provides a formal assumption under which we can eliminate the confounding bias by controlling for X.
- **Conditional Ignorability/ Exogeneity:** Suppose that treatment status D is independent of potential outcomes $Y(d)$ conditional on a set of covariates X; that is, for each d:

$$D \perp\!\!\!\perp Y(d) \mid X.$$

- You may wonder why the term "ignorability" is used. The distribution of $Y(d)$ depends only on X and not on D, so the latter is "ignorable".
- Note that the conventional name used in econometrics for the ignorability assumption is **conditional exogeneity**.

- We can learn about the causal effect of D by comparing outcomes across treated and control units who have identical characteristics $X = x$ under the conditional exogeneity assumption.
- The idea of comparing observations who have identical characteristics is the essence of the so-called *conditioning* strategy to learning causal effects.
- As conditioning approaches produce a different contrast for every potential value of X, we may also wish to average the contrasts at different values of X over the distribution of characteristics to produce a summary measure of the causal effects.

Overlap/Full Support

- The conditional probability of receiving treatment, *the propensity score*, plays an important role in this approach.
- **Overlap/Full Support:** The probability of receiving treatment given X, the propensity score $p(X) := \Pr(D = 1|X)$, is non-degenerate:

$$\Pr(0 < p(X) < 1) = 1.$$

- The overlap assumption requires that there is proper randomization or variation in D at each value x in the support of X.
- Without this condition, there are values x in the support of X where we cannot construct a contrast between treatment and control units.
- We cannot learn the conditional average treatment effect at these values of X and thus are also unable to learn the unconditional average effect of the treatment.

Conditioning on X Removes Selection Bias

Theorem 1 Under Conditional Ignorability and Overlap, the conditional expectation function of observed outcome Y given $D = d$ and X recovers the conditional expectation of the potential outcome $Y(d)$ given X:

$$\mathbb{E}[Y \mid D = d, X] = \mathbb{E}[Y(d) \mid D = d, X] = \mathbb{E}[Y(d) \mid X].$$

- To prove Theorem 1, note that the overlap assumption makes it possible to condition on the events $\{D = 0, X\}$ and $\{D = 1, X\}$ at any value in the support of X and that the second equality holds by conditional exogeneity/ignorability.

- Hence, the Conditional Average Predictive Effect (CAPE),

$$\pi(X) = \mathbb{E}[Y \mid D = 1, X] - \mathbb{E}[Y \mid D = 0, X],$$

is equal to the Conditional Average Treatment Effect (CATE),

$$\delta(X) = \mathbb{E}[Y(1) \mid X] - \mathbb{E}[Y(0) \mid X].$$

Thus, the APE and ATE also agree:

$$\delta = \mathbb{E}[\delta(X)] = \mathbb{E}[\pi(X)] = \pi.$$

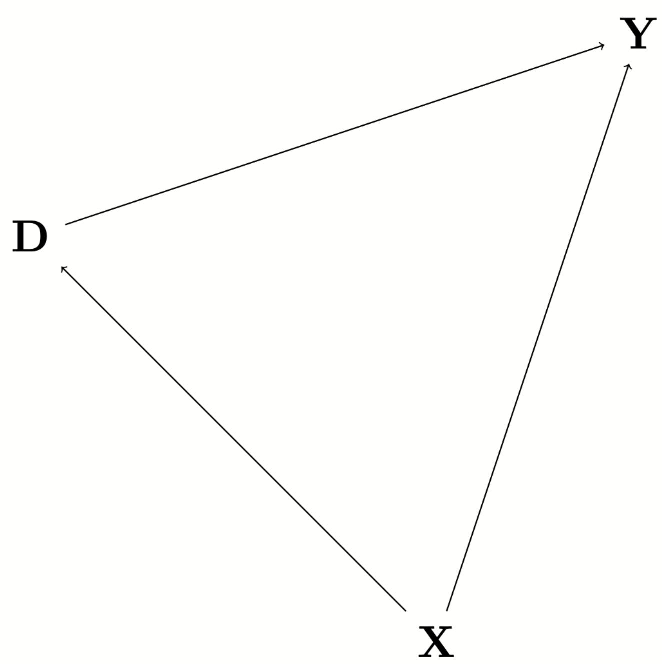
Conditional Exogeneity via Causal Diagrams

- It is possible to illustrate the previous set-up and assumptions graphically as follows:

$$
\begin{tikzcd}
\mathbf{D} & \mathbf{d} \arrow[r] & \mathbf{Y(d)} \\
 & \mathbf{X} \arrow[ul] \arrow[ur] &
\end{tikzcd}
$$

- In this graph, we show the potential outcome $Y(d)$ as a node and the potential treatment status d as another node. The latter node is deterministic.
- There is an arrow from d to $Y(d)$ indicating the dependency.
- The pre-treatment covariates X affect both the realized treatment variable D and the potential outcomes $Y(d)$, as shown by the arrow from X to D and from X to $Y(d)$.
- The assigned treatment variable D is independent of the node $d \mapsto Y(d)$, conditional on X, which is shown by the absence of the arrow between these two nodes.

- The potential outcome process $d \mapsto Y(d)$ and treatment assignment jointly determine the realized outcome variable Y via the assignment $Y := Y(D)$. This generates the following causal diagram:



- This graph says that X is generated first. D is then generated, with the distribution of D depending on X. Finally, Y is generated, with its distribution depending on both D and X. Here, after conditioning on X, the statistical dependence (association) between D and Y only reflects the causal channel, $D \to Y$ allowing us to uncover the ATE, for example.

Identification Using Propensity Scores

- The fact that conditioning on the right set of controls removes selection bias has long been recognized by researchers employing regression methods. Rosenbaum and Rubin made the much more subtle point that conditioning on only the propensity score

  $$p(X) = \Pr[D = 1 \mid X]$$

  also suffices to remove the selection bias.

Theorem 2 (Rosenbaum and Rubin: Conditioning on the Propensity Score Removes Selection Bias) Under Exogeneity and Overlap, D is generated independently of $Y(d)$ for each d, conditional on the propensity score $p(X)$: For each d,

$$D \perp\!\!\!\perp Y(d) \mid p(X).$$

- In other words, conditional on $p(X) = p$, variation in D is as good as randomly assigned. This fact makes $p(X)$ a "minimal sufficient" statistic, conditioning on which removes selection bias under ignorability.
- Conditioning on the propensity score rather than X itself is a useful empirical strategy when X is high dimensional and $p(X)$ is available (in stratified RCTs) or can be approximated accurately.
- An interesting example where the propensity score is not known but can be well-approximated is the examination in Abdulkadiroğlu et al. (2017) of the causal effect of attendance at a particular school or group of schools relative to one or more alternative schools (e.g., "elite" vs. "non-elite" schools) in settings where matching algorithms are used to assign students to schools. In this example, we can think of these student assignment mechanisms as $p(X)$.
- In such scenarios, we can simply use $p(X)$ as a control in place of the high dimensional set of characteristics, X, and thus bypass a potentially complicated high dimensional estimation problem.

Identification by Propensity Score Reweighting

- An alternative approach, known as the Horvitz-Thompson method (Horvitz and Thompson (1952)), uses propensity score reweighting to recover averages of potential outcomes.

Theorem 3 (Horvitz-Thompson: Propensity Score Reweighting Removes Bias) Under Conditional Exogeneity and Overlap, the conditional expectation of an appropriately reweighed observed outcome Y, given X, identifies the conditional average of potential outcome $Y(d)$ given X:

$$\mathbb{E}[Y 1(D = d)/\Pr(D = d|X) \mid X] = \mathbb{E}[Y(d) \mid X]$$

Then, averaging over X identifies the average potential outcome:

$$\mathbb{E}[Y 1(D = d)/\Pr(D = d|X)] = \mathbb{E}[Y(d)]$$

- To prove this result, note

$$
\begin{aligned}
&\mathbb{E}[Y1(D=d)/\Pr(D=d|X) \mid X] \\
&= \mathbb{E}[Y1(D=d) \mid X]/\Pr(D=d|X) \\
&= \mathbb{E}[Y(d) \mid X]\mathbb{E}[1(D=d) \mid X]/\Pr(D=d|X) \\
&= \mathbb{E}[Y(d) \mid X],
\end{aligned}
$$

where we used the conditional ignorability in the second equality.

- As a consequence, we can identify average treatment effects by simple averaging of transformed outcomes:

$$
\delta = \mathbb{E}[YH], \quad H = \frac{1(D=1)}{\Pr(D=1|X)} - \frac{1(D=0)}{\Pr(D=0|X)},
$$

  where H is called the Horvitz-Thompson transform.

- Note that this reduces to the difference of means in the control and treatment groups when the propensity score is constant.

Stratified RCTs

- **Generalized/Stratified RCT:** If under conditional exogeneity and consistency, the propensity score $p(X)$ is known, the setting is called a generalized or stratified RCT.
- In the case, we are essentially back to a classical RCT.
- Propensity score reweighting is generally not the most efficient approach to estimating treatment effects from statistical point of view because it ignores any dependence between the outcomes and controls, X, that is not captured by the propensity score.
- By exploiting dependence between the outcomes and X not captured by the propensity score, more efficient estimation of treatment can occur as using this dependence "de-noises" the outcome.
- Moreover, estimation based on only propensity score reweighting fails under imbalances that might arise due to imperfect data collection.

Covariate Balance Checks

- Given a propensity score $p(X)$, which is available in the stratified RCT settings outlined above, we can check if the RCT is valid (randomization is successful) by performing a **covariate balance check**:

$$\mathbb{E}[H \mid X] = 0.$$

- In a linear framework, this check can be done by running a regression of H on W, a dictionary of transformations of X, and testing if W predicts H. W predicting H suggests that the RCTs randomization protocol did not go as planned.
- In the reemployment experiment, we observed that the balance did not seem satisfied across age groups; specifically, we found more younger workers in the control group. Hence, further controlling for age makes sense and results in modest changes to estimates of the treatment effect.
- Of course, there is no guarantee that controlling for observed covariates can overcome selection bias in compromised RCTs in general because unobserved covariates may be driving the bias.

Average Treatment Effect for Groups and on the Treated

- In addition to unconditional average treatment effects (ATE) or average treatment effects at specific values of the covariates $X = x$, we may be interested in average effects within specific subpopulations.
- A leading example of an interesting subpopulation treatment effect is a group ATE (GATE):

$$\bar{\delta} = \mathbb{E}[Y(1) - Y(0)|G = 1]$$

  where G is a group indicator defined in terms of X's, such as $G = 1(X \in R)$ where R is a region.
- For example, we might be interested in the effects of a training program among younger people, say between 18 and 30 years old ($G = 1(18 \leq age \leq 30)$); among people older than 30 years old (so $G = 1(30 < age)$); and differences between these two groups.

- We can immediately obtain the GATE using the identification results above and law of iterated expectations:

$$
\begin{aligned}
\mathbb{E}[Y(1) - Y(0)|G = 1] &= \mathbb{E}[\mathbb{E}[Y|D = 1, X] - \mathbb{E}[Y|D = 0, X]|G = 1] \\
&= \mathbb{E}[HY|G = 1].
\end{aligned}
$$

- That is, we can identify GATEs either by taking the difference in regression functions or applying propensity score reweighting of outcomes and then averaging over group G.

Average Treatment Effect on the Treatet

- We next consider treatment effects for the subpopulation of treated units, the *average treatment effect on the treated* (ATET):

$$\delta_1 = \mathbb{E}[(Y(1) - Y(0) \mid D = 1].$$

- For example, consider training completion as treatment, D, and X a vector of pre-treatment variables such that unconfoundedness holds. Consider the question:

How much more do trainees earn on average after going through the training program?

- The ATET, δ_1, is the parameter that answers such questions about counterfactuals. The ATET is identified by

$$\mathbb{E}[\mathbb{E}[Y|D = 1, X] - \mathbb{E}[Y|D = 0, X] \mid D = 1].$$

Connection to Linear Regression

- Our tools can be used to perform statistical inference on ATEs.
- We briefly discuss how (high dimensional) regression can be used to retrieve causal estimates when conditional ignorability/exogeneity holds.
- The simplest instance of the problem is when the conditional expectation function of Y given D and X is linear,

  $$\mathbb{E}[Y \mid D, X] = \alpha D + \beta' W,$$

  which gives a model

  $$Y = \alpha D + \beta' W + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D, X] = 0.$$

- In this model, α identifies δ, $\delta = \alpha$, under the linearity assumption and exogeneity, and our inference tools for α automatically carry over to δ.

- Of course, the assumption of linearity is restrictive. A simple way to relax this is to consider interactions. One version of this approach takes all interactions and assumes

$$\mathbb{E}[Y \mid D, X] = \alpha_1 D + \alpha_2 DW + \beta_1 + \beta' W,$$

where we also maintain that we are working with centered covariates: $\mathbb{E}[W] = 0$.
- This model is still linear and results for linear models carry over to this case as well.
- We then recover the ATE as $\delta = \alpha_1$ and CATE as

$$\delta(X) = \alpha_1 + \alpha_2' W.$$

- We can use partialling out methods, such as OLS in low-dimensional case and Double Lasso (and variants) in high-dimensional case, to perform inference on α_1 and components of α_2, or even β_1.

What if the propensity score is known?

- When the propensity score $p(X)$ is known, we can include is as part of W in the formulations from the previous section.
- Another useful strategy is to consider regression models where HY is the dependent variable (Chernozhukov et al. (2018)):

$$\mathbb{E}[HY \mid H, X] = \alpha_1 + \alpha_2' W + \beta_1 H + \beta_2' HW.$$

- In this regression model, we recover the ATE as $\delta = \alpha_1$ and CATE as $\delta(X) = \alpha_1 + \alpha_2' W$.
- We can use partialling out methods, such as Double Lasso, to perform inference on α_1 and components of α_2.
- We also discuss inference on CATE using more general machine learning methods in later lectures.

Bibliography

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